A Fully Integer, Subject-Independent sEMG Gesture Classifier ASIC: Design Decisions, Training Results and Verification

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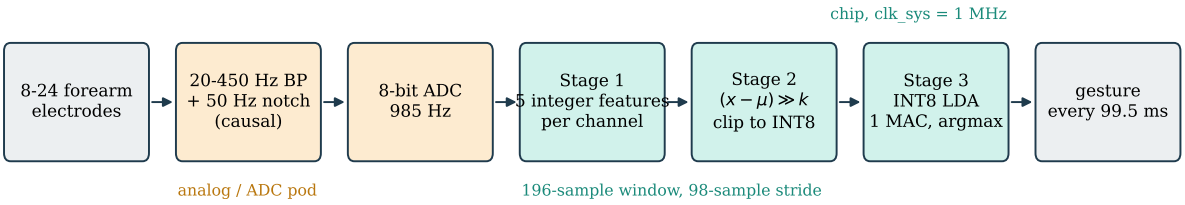


Figure 1: The classifier. Surface EMG is filtered and digitised in the electrode pods; the chip computes five integer features per channel, normalises them with a shift, and an INT8 LDA with one multiplier decides one of four gestures every 98 samples.

Abstract

We designed and verified a fixed-function digital core that classifies four hand and wrist gestures from 8 to 24 surface-EMG electrodes using only integer arithmetic and one 8×8 multiplier. Every design choice was settled by a subject-independent experiment on the public FORS-EMG dataset (leave-one-subject-out, paired statistics). The gesture set was the largest single factor (+14.1 pp). Classifier capacity did not matter: five MLP widths, logistic regression, a linear SVM and LDA are statistically tied, and INT8 quantisation costs 0.18 pp on average over 28 comparisons. Replacing the floating-point features with add, compare and shift operations and a shift normaliser costs no accuracy. A bit-exact golden model of the chip reaches 77.61±4.34% against 77.83±4.11% for the software INT8 LDA (no detectable difference), and the Verilog matches that model bit for bit at every channel count from 8 to 24, using at most 850 of the 1015 clock cycles available per sample.

1 Introduction

Wearable myoelectric control (prostheses, gesture interfaces) needs a classifier that runs continuously at very low power and works for a person it was never trained on.

- **Low power** favours integer arithmetic, few multiplications, small memories and a low clock rate (here 1 MHz).
- **Generalisation to a new user** is the hard part of sEMG classification: skin, fat, electrode placement and muscle physiology change the signal between people. We therefore report only subject-independent accuracy.

The work went through three phases: choosing the gesture set and classifier (Sections 3–5), making the front end integer-only (Sections 6–7), and designing and verifying the hardware (Sections 8–9). Section 10 collects every decision and its evidence in one table.

2 Dataset and evaluation protocol

Data. FORS-EMG: 19 subjects, 8 forearm electrodes, 985 Hz, 12 gestures in 3 forearm orientations (rest, supination, pronation), 5 trials each; every recording is 8000 samples (≈8.1 s). Figure 2 shows one recording of each gesture used on the chip.

Signal chain. A 4th-order Butterworth band-pass (20–450 Hz) and a 50 Hz notch ($Q = 30$), run causally as five biquads. The first experiments used zero-phase filtering, which hardware cannot run (it needs future samples), so all chip-related results use the causal filter. Figure 3 shows the effect: baseline drift below 20 Hz and the mains line are removed, and the 20–450 Hz band is kept.

Windows. 200 ms windows (196–197 samples) with a 100 ms stride.

Split. Leave-one-subject-out (LOSO): the held-out subject is never seen in training or normalisation. Within the training subjects, trial 5 is the validation set and trials 1–4 train. Windows are cut *after* the split. An early bug cut windows first; because neighbouring windows share half their samples, validation data leaked into training and produced a false 63% ceiling that did not depend on model capacity.

Statistics. Accuracy is the mean over held-out subjects with a 95% confidence interval. Two methods are compared per subject (paired): mean difference, its 95% CI, and the Wilcoxon signed-rank test (paired t-test in the earliest runs). A difference is called real only if its CI excludes zero.

The repository copy of the dataset lacks Subject 1, so runs made for this report use 18 subjects; runs made on the full dataset use 19. Every table states which.

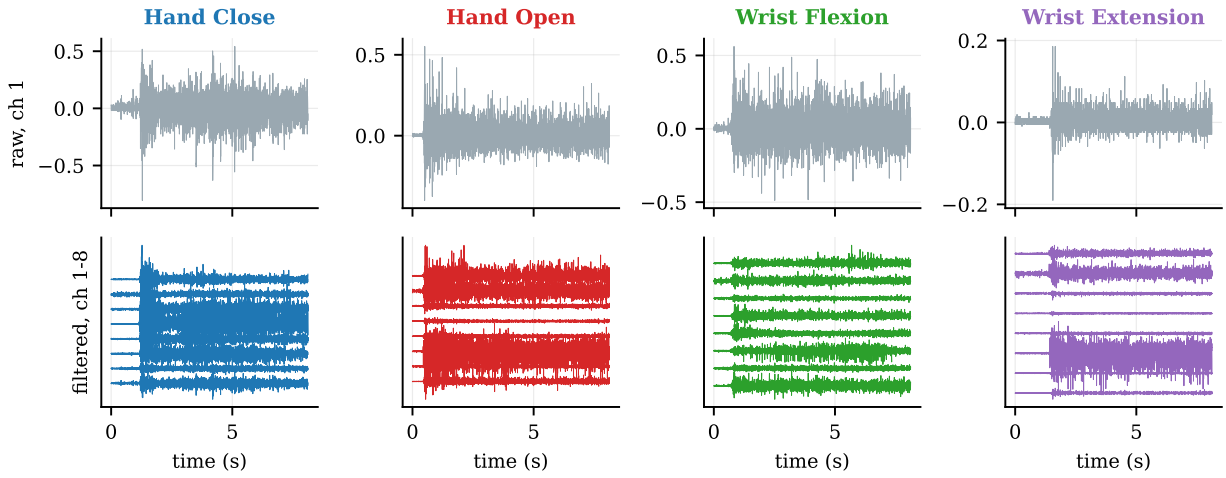


Figure 2: One recording per gesture (subject 2, rest orientation, trial 1). Top: raw channel 1. Bottom: all 8 channels after the causal filter, offset for clarity. The gestures differ in which channels are active, which is what the per-channel amplitude features capture.

3 Gesture set

The first set, {Hand Close, Hand Open, Index, Index Little} (Set A), stopped at about 65% whatever the model. Its confusion matrix showed that Index ↔ Index Little alone caused about 40% of all errors: the two finger movements recruit overlapping deep flexor compartments, which 8 surface electrodes cannot separate.

Decision. Use {Hand Close, Hand Open, Wrist Flexion, Wrist Extension} (Set B): gross movements driven by distinct, superficial muscles. This gained **+14.1 pp** (paired t-test $p = 3.8 \times 10^{-9}$, 95% CI $[+11.3, +16.9]$), more than any later model or hardware choice (Fig. 4).

Larger vocabularies cost accuracy (Table 1). The best 6 of the 12 gestures (Set D) were chosen by ranking all $\binom{12}{6} = 924$ subsets by their worst pairwise confusion in the 12-class confusion matrices. That ranking predicted 66.4% (band 65–75%) before training; the measured 69.0% fell inside it, so

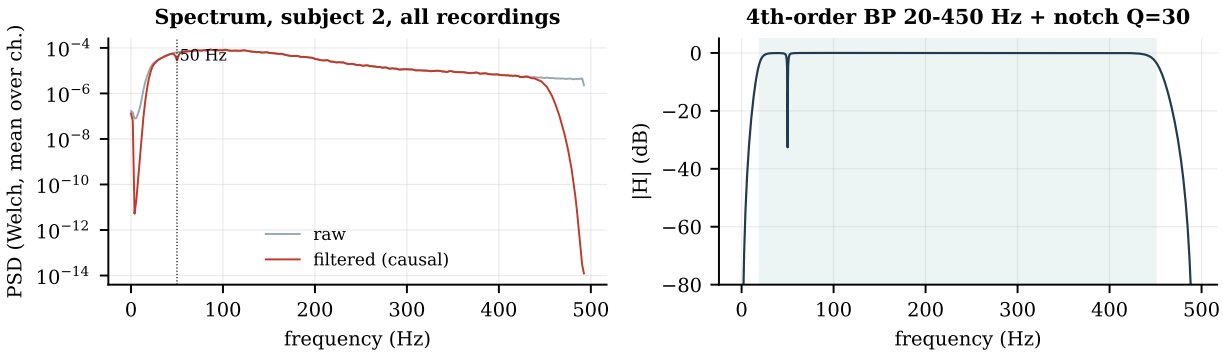


Figure 3: Left: power spectrum before and after the causal filter (subject 2, all recordings). Right: magnitude response of the filter.

existing confusion data can rank vocabularies without retraining. In Set D, Right Angle caused 44.5% of all errors (F1 0.583 against 0.63–0.78 for the others). In Set C (all 12), the MLP and logistic regression agreed on 9 of their 10 most confused pairs, so the errors come from the signals, not from the model.

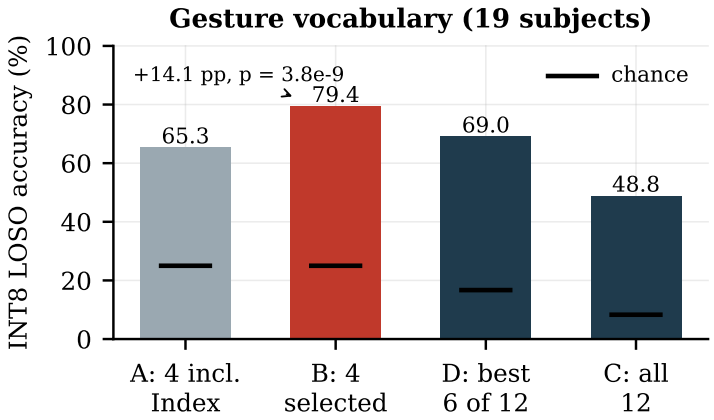


Figure 4: INT8 LOSO accuracy for each vocabulary (19 subjects); black bars: chance.

Table 1: Training results by vocabulary (INT8 and FP32, LOSO, 19 subjects).

Set	Model	FP32 %	INT8 %	Q-drop	MACs	Bytes
B (4)	MLP 64-16-4	79.69	79.38	0.32	1088	1168
D (6)	MLP 64-16-6	69.23	69.21	0.02	1120	1208
	Logistic reg.	68.95	68.74	0.21	384	408
C (12)	MLP 64-16-12	48.99	48.37	0.63	1216	1328
	Logistic reg.	49.71	49.28	0.43	768	816

Set B: ±8.20 (std over subjects). D tied $p = 0.49$; C tied $p = 0.197$.

4 Classifier

Width sweep (Set A). Five MLP widths, from a linear layer to 64-128-4, were trained under the same protocol three times, once per bug fix of the pipeline. Every time they were statistically indistinguishable: the spread between them (1.3 pp) was smaller than the ±5.9 pp CI of any one of them, and the ranking changed between runs (Table 3, Fig. 5).

Table 2: Per-class F1, Set B, MLP 64-16-4.

Hand Close	Hand Open	Wrist Flexion	Wrist Extension
0.838	0.695	0.883	0.749

Table 3: MLP width sweep, Set A (final pipeline, 19 subjects).

Architecture	FP32 %	INT8 %	Q-drop	Bytes	MACs
64-4 (linear)	65.17	65.28	−0.12	272	256
64-16-4	63.71	63.63	0.07	1168	1088
64-32-4	63.89	63.93	−0.04	2320	2176
64-64-4	64.80	64.87	−0.07	4624	4352
64-128-4	64.22	64.02	0.21	9232	8704

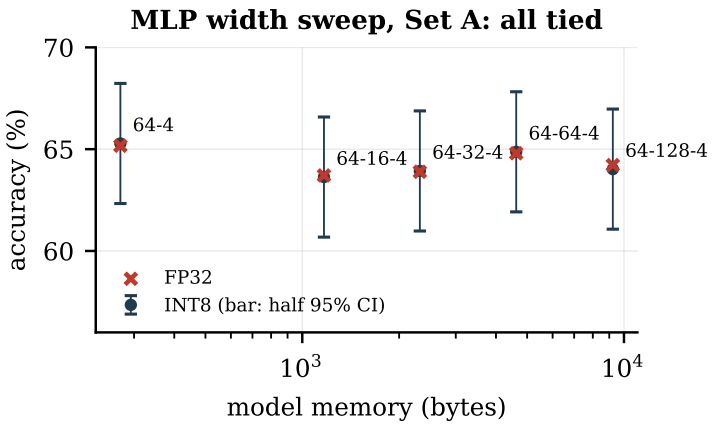


Figure 5: MLP width against memory, Set A. A 34× larger model is not more accurate.

Eight classifiers (Set B). Using 64 float features (8 types × 8 channels) and a 10 KB memory budget (Table 4, Fig. 6):

- the MLP, logistic regression, linear SVM and LDA are statistically tied;
- gradient boosting ties too, but needs 19 KB;
- random forest and extra trees are worse and over budget; the only tree inside the budget, a single decision tree, is 6.4 pp worse.

Table 4: Eight classifiers, Set B (LOSO, 19 subjects). "vs MLP": paired test.

Model	FP32	INT8	Q-drop	MACs	Bytes	vs MLP
MLP 64-16-4	79.65	79.57	0.08	1088	1168	–
Hist. grad. boost	79.63	79.26	0.37	0	19 200	tied (p=.78)
Logistic reg.	79.33	79.19	0.14	256	272	tied (p=.44)
Linear SVM	79.31	78.30	1.01	256	272	tied (p=.29)
LDA	77.91	77.87	0.04	256	272	tied (p=.09)
Random forest	77.84	77.43	0.41	0	53 145	worse (p=.046)
Extra trees	76.29	76.04	0.25	0	38 015	worse (p=.0009)
Decision tree	73.69	73.17	0.52	0	5055	worse (p=.0002)

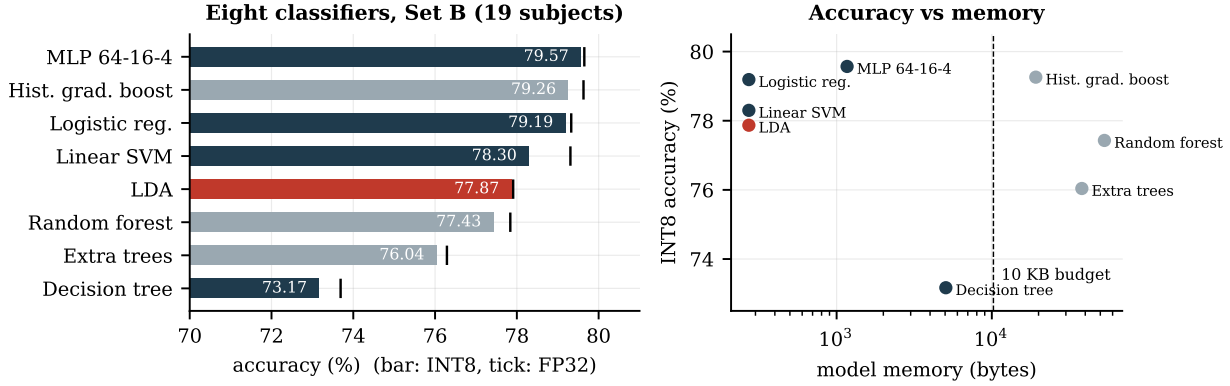


Figure 6: Left: INT8 accuracy (bar) and FP32 accuracy (tick) of the eight classifiers. Right: accuracy against memory; grey models break the 10 KB budget, red is the LDA used on the chip.

Finding: capacity is not the bottleneck. The limit is the information in 8 surface channels, not the model; this held at 4, 6 and 12 classes.

Decision: LDA for the chip. Logistic regression had the best point estimate among the cheap models. We still chose LDA, because:

1. its training is closed-form (class means and one shared covariance), with no iterative optimiser, so retraining for a new electrode layout is a few matrix operations;
2. its weights are stored, not hard-wired, so a retrained model is loaded through the configuration port without new silicon;
3. its scores have a probabilistic meaning (log-likelihoods up to a shared constant).

It costs 1.7 pp against the MLP, inside the tie ($p = 0.09$).

CNNs and a programmable accelerator. Yash and Digvijay trained raw-signal CNNs (depthwise-separable and two-branch designs, batch normalisation folded into the weights, INT8) and co-designed an 8-bit ISA for a programmable MAC accelerator. In October 2026 the scope was fixed to the LDA: capacity gave no gain on hand-crafted features, a CNN needs many more MACs and weights, and a programmable controller adds control logic and verification effort the 4-gesture task does not need.

5 INT8 quantisation

Weights and activations are symmetric per-tensor INT8 with the scale set at the 99.9th percentile of $|t|$; biases are INT32 on the scale $s_W s_x$, so they add straight into the accumulator. All scales come from the training subjects only.

Result. Over 28 FP32-vs-INT8 comparisons (every model and gesture set), the loss was 0.18 pp on average (median 0.11, standard deviation 0.25, worst 1.01 for the linear SVM, best −0.16). 25 of 28 losses were under 0.5 pp and 27 of 28 under 1 pp (Fig. 7). The linear SVM is the outlier because its hinge loss produces larger weights near the boundary.

A requantisation bug (a shift of $\lceil -\log_2 M \rceil$, undefined when the multiplier $M > 1$) would have cost over 50 pp; a synthetic unit test caught it before any result was reported. The fixed form is $y = (z\,M_0 + 2^{30}) \gg 31$ with $M_0 = \mathrm{round}(M\,2^{31})$, valid for any $M > 0$.

No requantisation on the chip. The LDA only needs arg max. If every class score shares one positive scale $s > 0$, then

$$\arg\max_k s\,S_k = \arg\max_k S_k,$$

so the chip keeps one INT32 accumulator per class and never rescales. This requires one weight scale for all classes.

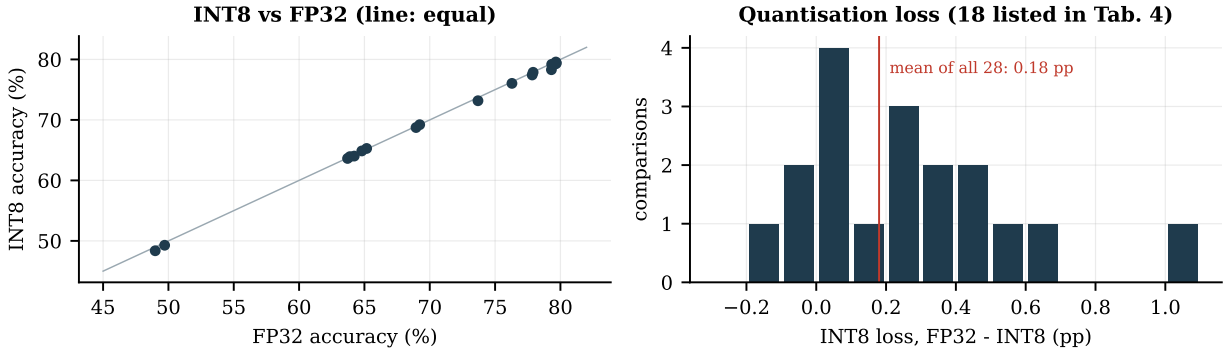


Figure 7: INT8 against FP32 for the 18 results listed in Tables 1–4 (left) and the distribution of their losses (right). INT8 is lossless for practical purposes.

6 Feature-type selection

The 64 float features used 8 types per channel. To cut the input size we selected types by nested LOSO on the INT8 LDA (18 subjects): for each held-out subject, all 127 subsets of 7 candidate types were scored by an inner LOSO on the other 17, and the smallest subset within one paired standard error of the best was kept. IEMG was excluded beforehand, since IEMG = L·MAV gives an identical LDA.

Redundancy. MAV, RMS, WL and VAR are almost the same signal (Spearman $|\rho| \geq 0.977$, Fig. 8); ZC, SSC and WAMP carry different information. WAMP was saturated (98% of sample differences exceed its threshold).

Result. {MAV, WL, ZC, SSC, VAR} was selected overall. MAV, ZC and VAR were chosen in all 18 folds, WL in 17 and SSC in 14. Accuracy saturates at $k = 5$ (Fig. 8). The selected set reaches **77.08±4.97%** against 77.31±5.02% with all 8 types ($\Delta = -0.23 \pm 0.42$ pp, $p = 0.20$, non-inferior at a −1 pp margin), with 40 inputs instead of 64: 176 instead of 272 bytes and 160 instead of 256 MACs. Table 5 lists every fold.

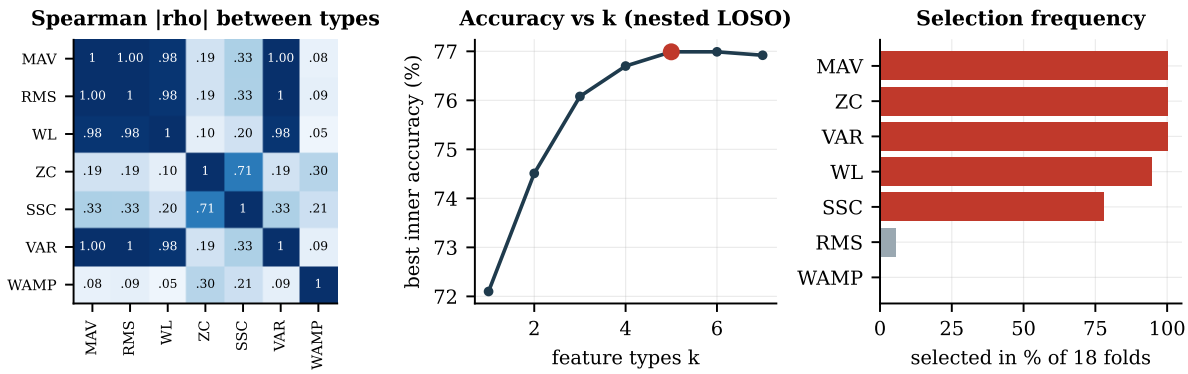


Figure 8: Feature-type selection. Left: rank correlation between types (mean over channels). Middle: best inner-LOSO accuracy for each number of types. Right: how often each type was selected over the 18 outer folds.

Table 5: Nested LOSO feature selection, per held-out subject (INT8 %).

Test	Selected types	Inner	Selected	All 8
S02	MAV WL ZC SSC VAR	77.10	68.90	68.92
S03	MAV WL ZC SSC VAR	78.87	65.00	64.94
S04	MAV WL ZC SSC VAR	76.64	76.77	75.17
S05	MAV WL ZC VAR	77.61	67.75	68.75
S06	MAV WL ZC SSC VAR	76.29	87.33	87.98
S07	MAV WL ZC SSC VAR	77.17	71.12	70.46
S08	MAV WL ZC SSC VAR	76.61	81.12	81.50
S09	MAV WL ZC VAR	76.32	80.12	80.40
S10	MAV WL ZC SSC VAR	76.65	81.71	81.75
S11	MAV WL ZC SSC VAR	76.43	85.69	85.73
S12	MAV WL ZC SSC VAR	76.51	82.54	83.06
S13	MAV WL ZC SSC VAR	78.00	63.08	63.33
S14	MAV WL ZC SSC VAR	76.57	84.10	84.31
S15	MAV WL ZC VAR	78.27	57.52	57.04
S16	MAV WL ZC VAR	76.93	70.75	72.02
S17	MAV WL ZC SSC VAR	76.22	91.79	91.94
S18	MAV RMS ZC SSC VAR	76.24	80.81	83.27
S19	MAV WL ZC SSC VAR	76.23	91.40	91.02
Mean			**77.08**	77.31

7 Integer front end

The selected float features need division, square roots and floating-point thresholds. We rebuilt them from add, subtract, absolute value, compare, shift and a leading-one detector.

- **SABS** $= \sum |x|$ replaces MAV. The $1/N$ is absorbed by normalisation, since LDA is invariant to rescaling one feature.
- **WL** $= \sum |x_n - x_{n-1}|$.
- **ZC**: sign-bit XOR of consecutive samples, with no amplitude dead-band.
- **SSC**: strict sign change of consecutive differences; only the sign of the previous difference is stored, so each channel's history needs 10 bits.
- **POW** replaces VAR: $\sum x^2$ with Mitchell's one-step square,

$$a^2 \approx a \cdot 2^{p+1} - 2^{2p}, \qquad p = \lfloor \log_2 a \rfloor,$$

one leading-one detector, two shifts and one subtraction. It never overestimates (Fig. 9); the summed feature has correlation $r = 0.9999$ with the exact sum.

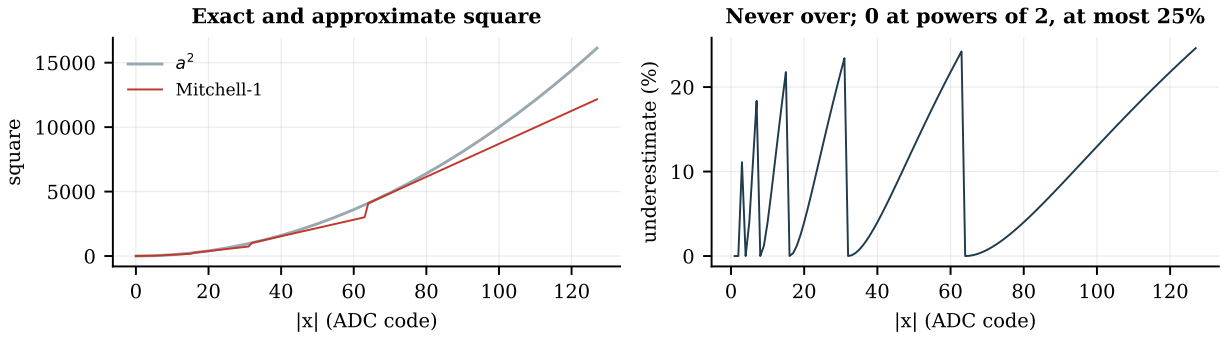


Figure 9: Mitchell-1 square of an 8-bit magnitude. The per-sample error is up to 25%, but it is a smooth, monotone distortion, and the window sum stays almost perfectly correlated with $\sum x^2$.

Normaliser. Z-scoring and float input quantisation are replaced by

$$q_f = \mathrm{clip}\big(((x_f - \mu_f) + 2^{k_f - 1}) \gg k_f,\ -128,\ 127\big),$$

with integer μ_f (rounded training mean) and the smallest $k_f \geq 1$ that never clips on the training subjects. Figure 10 shows the resulting INT8 inputs on held-out subjects and the shifts of one fold: count features need $k = 1$, amplitude features 4–6, power 7–12.

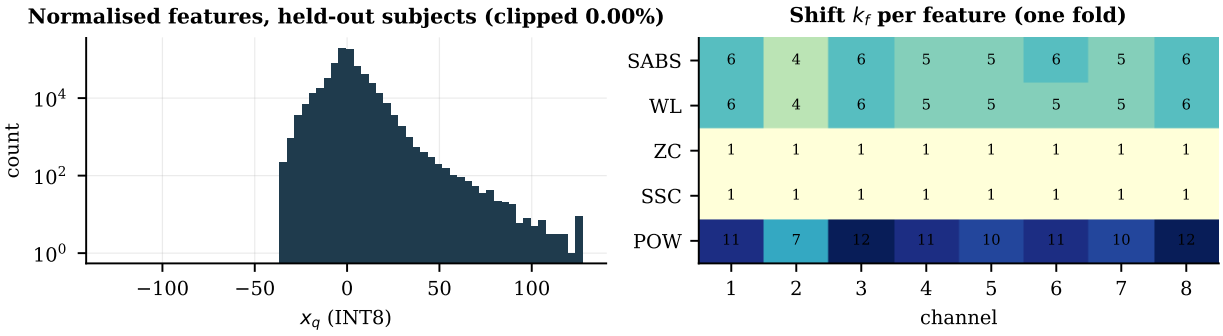


Figure 10: Left: distribution of x_q over all held-out windows (log scale). Right: k_f for each feature and channel in one fold.

Result. Six integer variants were trained against the float reference R on the same folds (Table 6, Fig. 11). Feature fidelity on all windows: Mitchell-1 POW $r = 0.9999$ with the exact sum, Mitchell-2 $r = 1$, $(\sum |x|)^2$ $r = 0.994$, ZC without dead-band $r = 0.9987$ with the float ZC, SABS $r = 1$ with MAV.

- every variant that keeps a power feature is non-inferior to R (margin -1 pp);
- dropping power (H5) loses about 2 pp and fails, so a power feature is required;
- the shift normaliser costs nothing against the standard one (all six pass).

Decision: H6, the cheapest variant, with the shift normaliser. On 19 subjects this fully integer front end gives 77.79±4.39% against 77.88% for the float pipeline.

Table 6: Integer feature sets, INT8 LDA, LOSO (18 subjects). Δ: paired difference to R (standard normaliser) with the shift normaliser, 95% CI.

	Power / ZC	Train	Std norm.	Shift norm.	Δ vs R (pp)
R	float reference	81.56	77.45	–	–
H1	exact $\sum x^2$	81.51	77.33	77.63	+0.19 [−0.36, +0.73]
H2	Mitchell-2	81.52	77.29	77.56	+0.11 [−0.41, +0.63]
H3	Mitchell-1	81.50	77.26	77.39	−0.05 [−0.57, +0.47]
H4	$(\sum \lvert x \rvert)^2$	81.34	77.10	77.14	−0.31 [−0.88, +0.27]
H5	no power	79.41	75.38	75.54	−1.91 [−2.94, −0.87]
H6	Mitchell-1, ZC no db	81.51	77.29	77.39	−0.05 [−0.59, +0.48]

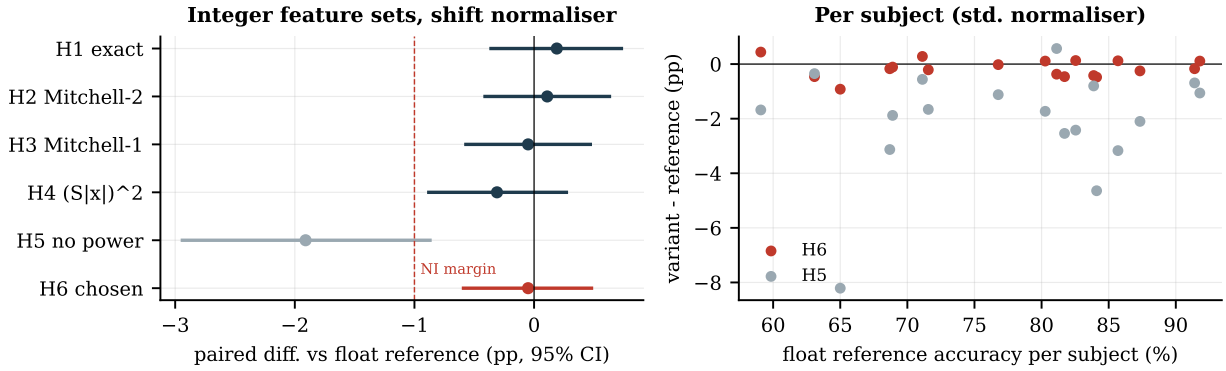


Figure 11: Left: paired difference of each integer variant to the float reference (shift normaliser, 95% CI); the dashed line is the non-inferiority margin. Right: per-subject difference for the chosen H6 and for H5, which drops the power feature.

8 Hardware architecture

8.1 System partition

We considered three ways to support 8 to 24 electrodes: several compute boards exchanging partial LDA scores (exact, since the score is a sum over features), master/slave boards sharing state, or one compute core with pluggable acquisition pods. At 1 MHz the features take a few cycles per channel per sample, so one core handles up to 40 channels within the 1000-cycle sample period, and the LDA needs $6(D + 1)$ cycles per decision. Distributing the computation saves nothing and adds synchronisation. **Decision**: one compute core plus 1–3 daisy-chained 8-channel ADC pods, with the channel count set by one register (Fig. 12).

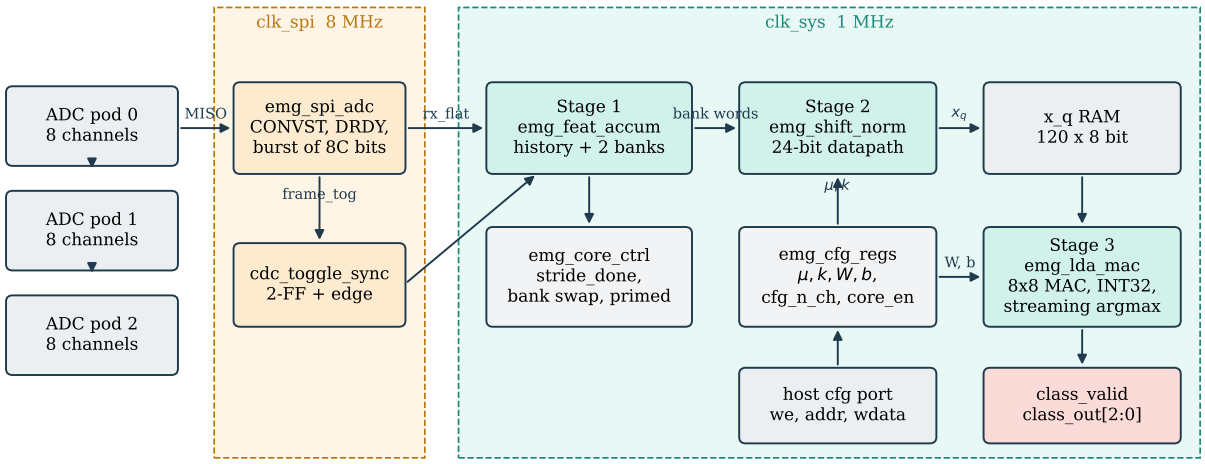


Figure 12: Chip architecture with the RTL module names. Orange: 8 MHz ADC-interface domain. Green: 1 MHz datapath. Only the frame event crosses between the two.

8.2 Datapath

1. **Stage 1, features** (`emg_feat_accum`). For every frame of C samples, a two-stage pipeline reads the channel's history and bank word from RAM, computes the five features combinationally and writes the sums back: C cycles per sample. Two ping-pong banks of 98 samples give the 196-sample window with a 98-sample stride and a single write pointer (Fig. 13).
2. **Stage 2, normaliser** (`emg_shift_norm`). After every 98th sample it adds the two bank fields, subtracts μ_f, rounds, shifts by k_f and clips: $D + 2$ cycles, 24-bit signed datapath.
3. **Stage 3, LDA** (`emg_lda_mac`). One signed 8×8 multiplier and an INT32 accumulator. Per class it loads the bias, then performs D MACs, with one weight-address counter for all classes. A streaming argmax with strict $>$ (ties keep the lower class) replaces a comparator tree. Total $6(D + 1) + 2$ cycles; unused class slots get bias -2^{31}.

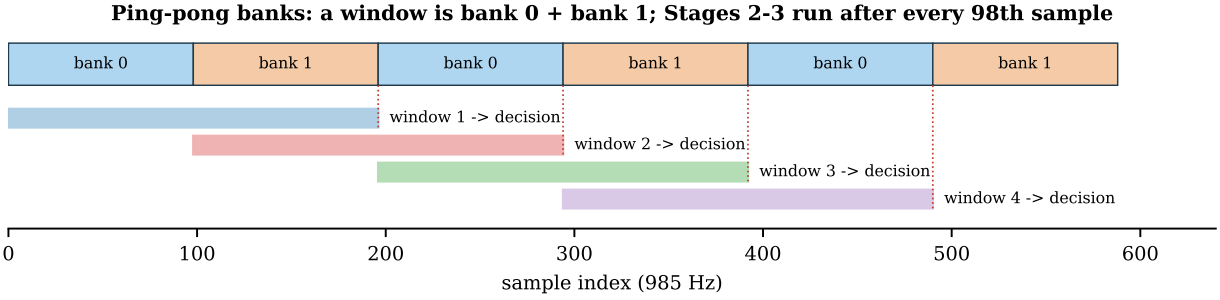


Figure 13: Windowing with two banks. Each bank accumulates 98 samples; a window is the sum of the last two banks, so each new bank completes a window and triggers one decision.

8.3 Clocking, crossing and reset

The ADC interface runs on an 8 MHz `clk_spi`. It generates CONVST every 8120 cycles (985 Hz), detects the fall of DRDY through a 2-flip-flop synchroniser, and reads $8C$ bits at 4 MHz. Only one event crosses into the core: a frame toggle, through a 2-flip-flop toggle synchroniser. The data crosses unsynchronised, because it is stable for about 1 ms around the event; in static timing analysis it is

a multicycle path. Each domain has its own reset synchroniser (asynchronous assert, synchronous release).

8.4 Timing and memory

Stage 1 costs C cycles on every sample. Stages 2 and 3 run once per 98 samples, and their measured cost grows by 35 cycles per channel, from 290 to 850 (Fig. 14). Even at 24 channels everything finishes within one 1015-cycle sample period, so one multiplier suffices.

All constants live in writable memories, not ROM, because every channel count and electrode layout needs its own model. The total is 13.5 kbit, sized for 24 channels and 6 classes; every memory has a 1-cycle synchronous read, so the RTL maps to SRAM macros or to flip-flops unchanged.

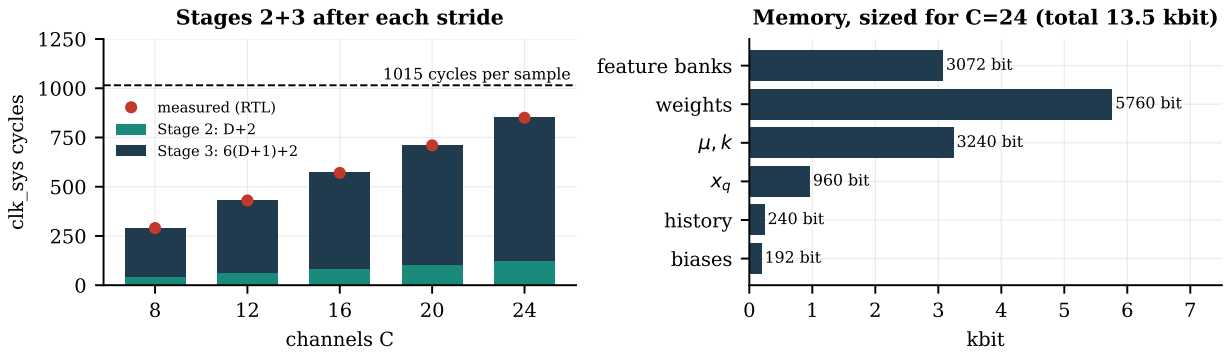


Figure 14: Left: Stage 2 + 3 cycles per decision, formula (bars) and RTL simulation (dots). Right: memory by block.

9 Verification

Golden model. `lda_golden_model.py` reproduces every integer of Stages 1–3 in two forms: a sample-by-sample model that mirrors the RTL's counters, and a vectorised model used for training. The two are cross-checked on every run (81 decisions on a real recording). Training uses the model's own bit-exact features, so training and hardware see identical numbers.

Does the chip arithmetic cost accuracy? The golden model (8-bit ADC, streamed 196-sample windows, shift normaliser, single weight scale) was trained and tested by LOSO and compared with the software INT8 LDA on the same folds (Table 7, Fig. 15): 77.61±4.34% against 77.83±4.11%, $\Delta = -0.21$ pp, 95% CI $[-0.72, +0.30]$, $p = 0.49$, 9 subjects up and 9 down. **No detectable difference.**

The value ranges on real data leave wide margins: μ_f uses at most 13 of its 22 bits, k_f is 1–12, $|S| \leq 4047$ against the INT32 limit, $|b| \leq 819$, and no feature clips.

Where the errors are. The confusion matrix of the chip model (Fig. 16) shows that most errors are Wrist Extension predicted as Hand Open (29%) and the reverse (18%), the pair that shares extensor-digitorum activity. Hand Close and Wrist Flexion are recognised in 82–85% of windows. The LDA projection (Fig. 17) shows the same: Hand Open and Wrist Extension overlap, and the held-out subject sits close to the training clusters.

Unit tests. The feature ALU was tested exhaustively on all 196 608 input combinations (every pair of INT8 values with the three signs of the previous difference). Each stage had its own testbench. We checked that the testbenches catch real bugs by inserting deliberate ones: a logical instead of arithmetic shift, a missing rounding term, a wrong field, an unsigned multiply, $\geq$ instead of $>$, a dropped bias and a too-narrow accumulator. All were caught.

Chip-level test. A cocotb testbench (Fig. 18) trains a model with the golden model, loads it through the configuration port, drives an ADC daisy-chain model with a gesture stream at the real clock rates, and at every stride compares the D values of x_q, the six class scores and the decided class

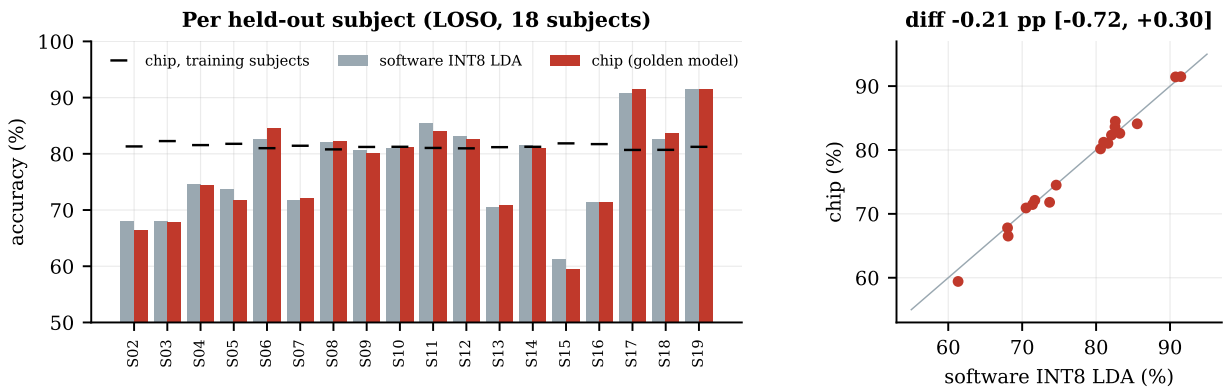


Figure 15: Left: held-out accuracy of the chip model and the software INT8 LDA for each subject; dashes show the chip model's accuracy on its own training subjects. Right: the two per subject; points lie on the diagonal.

Table 7: Chip model vs software INT8 LDA, per held-out subject (%).

Subj.	Chip	Software	Subj.	Chip	Software
S02	66.52	68.10	S11	84.10	85.56
S03	67.81	68.02	S12	82.60	83.21
S04	74.50	74.60	S13	70.92	70.52
S05	71.81	73.71	S14	81.04	81.60
S06	84.50	82.60	S15	59.42	61.33
S07	72.12	71.71	S16	71.46	71.40
S08	82.33	82.08	S17	91.44	90.75
S09	80.17	80.60	S18	83.62	82.56
S10	81.21	81.02	S19	91.48	91.46
Mean: chip **77.61**±4.34, software 77.83±4.11					

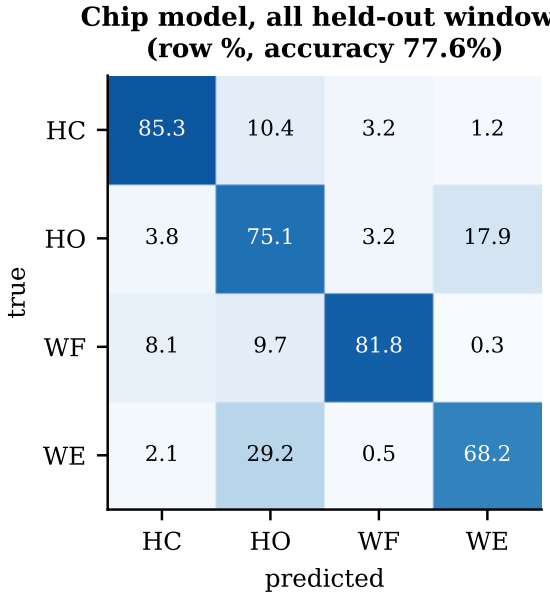


Figure 16: Confusion matrix of the chip model over all held-out windows of all 18 folds (86 400 windows).

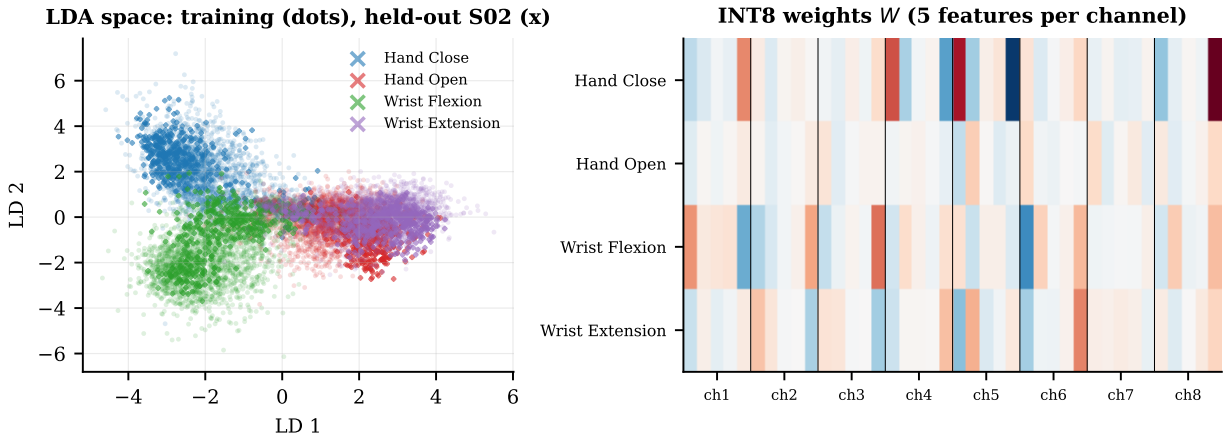


Figure 17: Left: first two discriminant directions of one fold, training windows (dots) and the held-out subject (crosses). Right: the INT8 weights loaded into the chip for that fold.

against the golden model. It then checks that the first decision is masked and that no overrun occurs. A failure names the first stage that disagrees. The design passes bit-exactly at $C = 8, 12, 16, 20, 24$ and lints cleanly in Icarus (`-Wall`) and Yosys.

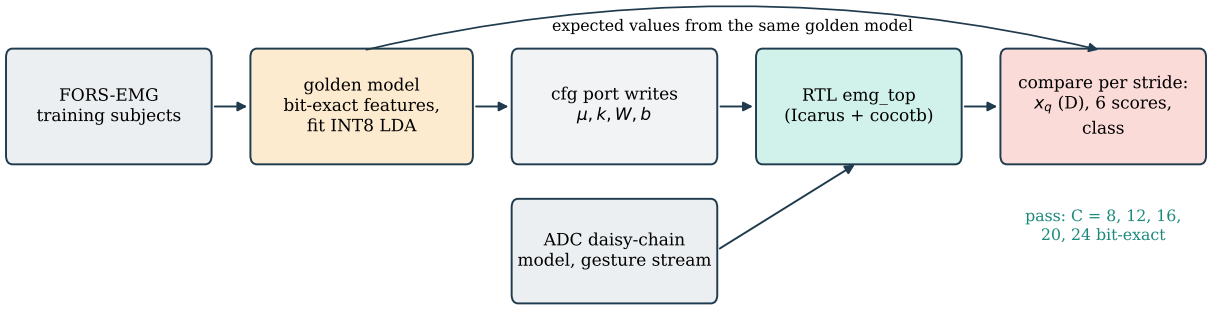


Figure 18: Chip-level verification: the same golden model trains the chip and produces the expected value of every intermediate signal.

10 Decisions and trade-offs

Table 8 lists every decision. Three trade-offs shaped the design most.

1. **Accuracy vs vocabulary.** Each added gesture costs accuracy. We favoured a reliable 4-gesture interface over a 12-gesture one at 49%.
2. **Model capacity vs cost.** A larger model gains nothing measurable over the LDA but multiplies MACs and memory.
3. **Generality vs area.** One core with a channel-count register serves 8 to 24 electrodes; the cost is sizing every memory for 24 channels.

11 Conclusion and future work

Gesture selection mattered most, model capacity did not, integer arithmetic and INT8 cost almost nothing, and a one-multiplier core at 1 MHz is enough for 24 electrodes. The RTL is verified bit-exactly against a model whose accuracy matches the software reference. Next steps:

1. synthesise in a target process to report area and power;

Table 8: Every major decision, the alternatives considered, and the evidence.

Decision	Chosen	Alternatives	Reason / evidence
Evaluation	LOSO, trial-level validation, windows cut after the split	window-level split; per-subject training	only LOSO measures a new user; the window-level split leaked data (false 63% ceiling)
Gesture set	HC, HO, WF, WE	Index/Index Little set; 6 or 12 gestures	+14.1 pp; finger gestures inseparable with 8 surface electrodes; 6 gestures 69.0%
Classifier	INT8 LDA	MLP, LogReg, SVM, trees, CNN	all cheap linear models tied with the MLP; LDA has closed-form training and probabilistic scores; trees over budget or worse
Arithmetic	INT8 weights, INT32 accumulator, no requantisation	FP32; per-class scales	0.18 pp mean loss; a shared scale keeps the argmax
Feature types	5 of 8	all 8; other subsets	nested LOSO, non-inferior, 40 instead of 64 inputs
Features	SABS, WL, ZC, SSC, Mitchell POW	float MAV/VAR, exact squares, log features	integer set matches the float reference; Mitchell square needs no multiplier
Normaliser	$(x - \mu) \gg k$, integer	z-score + float quantiser	no accuracy loss; only subtract, add and shift
Partition	1 compute core + 1–3 ADC pods	multi-board partial sums; master/slave	one core handles up to 40 channels at 1 MHz; distributing adds synchronisation for no gain
Stage 3	1 MAC, streaming argmax	parallel MACs; comparator tree	850 of 1015 cycles at $C = 24$
Windows	2 ping-pong banks of 98	sliding sums; 2 write pointers	one write pointer; window = bank 0 + bank 1
Memories	writable RAM, 1-cycle read	ROM; flip-flops with combinational read	each channel count needs its own model; maps to SRAM or flip-flops
Clock crossing	toggle synchroniser for the event only	synchronise the data	data stable for about 1 ms; one bit crosses
Verification	bit-exact golden model, unit and chip tests, injected bugs	accuracy-only checks	every intermediate value checked; injected bugs caught

2. test the core with real ADC pods and recorded EMG;

3. reduce the Hand Open / Wrist Extension confusion, the main remaining error.

A Reproducing the results

All commands run from the repository root; each script prints the tables quoted above. The figures are drawn by `python docs/report/make_figures.py`.

Result	Script	Subjects
Gesture sets, classifiers, INT8 loss	`analysis/classifier_classification.py`, `K_4_model_MLP.py`, `K_6_model_LR_comparison.py`	19
Feature-type selection	`analysis/fors_emg_feature_selection_lda.py`	18
Integer features H1–H6	`analysis/K_4_LDA_HW_Experiments.py`	18
Chip model vs software LDA	`golden_model/K_4_LDA_Golden_LOSO.py`	18
RTL	`RTL/tb/test_emg_top.py` (set `CHANNELS`)	synthetic

B Script outputs

Verbatim excerpts of the script printouts (18 subjects).

```
== golden_model/K_4_LDA_Golden_LOSO.py
--------------------------------------------------------------------------------
Model                    Test %
-----------------------  ------------
Chip LDA (golden model)  77.61 +-4.34
Validated INT8 LDA       77.83 +-4.11

Comparison           d pp  95% CI            Wilcoxon p  Up/Down    Verdict
-----------------  ------  --------------  ------------  ---------  -------------------------
chip vs validated   -0.21  [-0.72, +0.30]         0.486  9/9        no detectable difference

--------------------------------------------------------------------------------

== analysis/K_4_LDA_HW_Experiments.py
SABS vs float MAV              1

--------------------------------------------------------------------------------
  Accuracy (mean over 18 held-out subjects; +- = 95% CI)
--------------------------------------------------------------------------------
    Configuration                  Train INT8 %    Test FP32 %  Test INT8 % (std norm)    Test INT8 % (shift norm)
--  -----------------------------  --------------  ------------  ----------------------  --------------------------
R   Float reference                         81.56         77.27  77.45 +-4.89
H1  Exact Sum x^2 (serial shift-add)         81.51          77.2   77.33 +-4.89              77.63 +-4.67
H2  Mitchell-2 Sum x^2                       81.52          77.2   77.29 +-4.93              77.56 +-4.70
H3  Mitchell-1 Sum x^2                       81.5           77.2   77.26 +-4.92              77.39 +-4.68
H4  (Sum|x|)^2 per window                   81.34         76.96  77.10 +-4.95             77.14 +-4.72
H5  No power term                           79.41         75.43  75.38 +-5.16             75.54 +-5.06
H6  ZC no deadband + Mitchell-1             81.51         77.21  77.29 +-4.89             77.39 +-4.69

--------------------------------------------------------------------------------
  Paired comparisons, test INT8 (non-inferiority margin -1 pp)
--------------------------------------------------------------------------------
               Question                       d pp  95% CI            Wilcoxon p  Up/Down    NI
---------------  -------------------------  ------  --------------  ------------  ---------  ----
H1 vs R          integer features (std norm)   -0.12  [-0.27, +0.03]         0.039  5/12       PASS
H2 vs R          integer features (std norm)   -0.16  [-0.30, -0.01]         0.028  6/12       PASS

== analysis/fors_emg_feature_selection_lda.py (stage 4)
------------------------------------------------------------------------
Top-10 subsets                      k  Inner acc      Deficit pp    paired SE pp
---------------------------------  ---  ------------  ------------  --------------  ----------
{MAV, WL, ZC, SSC, VAR}             5  77.19%                0               0     <- selected
{MAV, WL, ZC, SSC, VAR, WAMP}       6  77.10%                0.09            0.08
{MAV, RMS, WL, ZC, SSC, VAR}        6  77.09%                0.11            0.14
{MAV, RMS, WL, ZC, SSC, VAR, WAMP}  7  77.04%                0.15            0.17
{MAV, WL, ZC, VAR}                  4  76.77%                0.42            0.25
{MAV, RMS, WL, ZC, VAR}             5  76.74%                0.45            0.23
{MAV, WL, ZC, VAR, WAMP}            5  76.72%                0.47            0.25
{MAV, RMS, WL, ZC, VAR, WAMP}       6  76.66%                0.53            0.24
{RMS, WL, ZC, SSC, VAR}             5  76.55%                0.65            0.24
{RMS, WL, ZC, SSC, VAR, WAMP}       6  76.53%                0.67            0.25

  * Final feature types : {MAV, WL, ZC, SSC, VAR}  (k = 5 of 8)
    Input dim           : 40  (was 64)
    LDA param bytes     : 176 (was 272)   MACs: 160 (was 256)
    Expected accuracy on a new subject (nested INT8): 77.08 +-4.97 %
```